
Extending an Induced Star Along a Geodesic

A Proof of Graffiti.pc Conjecture 141

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Abstract

For a finite connected simple graph G , let $g(G)$ be its girth, with $g(G)=0$ when G is acyclic. Let

$$\ell(v)=\alpha(G[N(v)]), \lambda(G)=\max_{(v \in V(G))} \ell(v),$$

and let $\text{tree}(G)$ denote the largest number of vertices in an induced tree of G . Graffiti.pc Conjecture 141 asserts

$$\text{tree}(G) \geq \lfloor g(G)/2 \rfloor - 1 + \lambda(G).$$

We prove the conjecture by a direct construction. When $g(G) \geq 4$, every open neighborhood is independent and hence $\lambda(G) = \Delta(G)$. Starting with the induced star at a maximum-degree vertex, we extend one branch along a geodesic of length $\lfloor g(G)/2 \rfloor - 1$. A breadth-first spanning-tree argument guarantees that such a geodesic exists, while the girth condition forbids every possible extra edge in the proposed induced tree. The proof has also been formalized in Lean 4 against mathlib; the formal theorem has exactly the statement used by the Formal Conjectures project.

1. Introduction

Graffiti.pc is a computer-assisted conjecture-making program in graph theory. Conjecture 141, recorded in *Written on the Wall II*, relates three quantities that are global, local, and metric in character: the order of a largest induced tree, the maximum independence number of a vertex neighborhood, and the girth.

The argument below is constructive. It identifies an explicit induced tree: retain every edge of a largest induced star, then continue one of its branches along a shortest path. There are two points to check. First, the graph must contain a vertex far enough from the center of the star. This follows from a breadth-first spanning tree: if all vertices were too close to the root, any edge outside that tree would close a cycle shorter than the girth. Second, no other leaf of the star can have an edge to the extended branch, because such an edge would again close a short cycle.

2. Definitions and statement

All graphs in this note are finite and simple. For a graph G and $v \in V(G)$, write $N(v)$ for the open neighborhood of v and $G[N(v)]$ for the graph it induces. Define

$$\ell(v) = \alpha(G[N(v)]), \lambda(G) = \max_{v \in V(G)} \ell(v).$$

The girth $g(G)$ is the length of a shortest cycle; following the convention of the conjecture, $g(G)=0$ if G is acyclic. Finally, $\text{tree}(G)$ is the maximum cardinality of a set $U \subseteq V(G)$ for which $G[U]$ is a tree.

Theorem (Graffiti.pc Conjecture 141). If G is a finite connected simple graph, then

$$\text{tree}(G) \geq \lfloor g(G)/2 \rfloor - 1 + \lambda(G).$$

3. The breadth-first distance lemma

We isolate the only metric ingredient.

Lemma 1. Let G be connected and contain a cycle, and let v be any vertex. If $g=g(G) \geq 4$ and

$$r = \lfloor g/2 \rfloor - 1,$$

then some vertex w satisfies $d_G(v, w) \geq r$.

Proof. Suppose instead that every vertex has distance at most $r-1$ from v . Take a breadth-first spanning tree T rooted at v . Because G has a cycle, T omits some edge xy of G . The unique x - y path in T , together with xy , is a cycle. Its length is at most

$$d_T(x, v) + d_T(v, y) + 1 = d_G(x, v) + d_G(v, y) + 1 \leq 2(r-1) + 1 = 2r-1 < g,$$

contrary to the definition of girth. Therefore such a vertex w exists. $\square$

The equality between the T -distance and the G -distance from the root is the defining property of a breadth-first spanning tree.

4. Proof of the conjecture

We divide the proof into three cases.

4.1. Acyclic graphs

If G is acyclic, then $g=0$. For any vertex v , choose an independent set $S \subseteq N(v)$ of cardinality $\ell(v)$. The set $v \cup S$ induces a star, and hence

$$\text{tree}(G) \geq 1 + \lambda(G).$$

This is more than enough because the conjectured right-hand side is $-1 + \lambda(G)$.

4.2. Girth three

The same induced-star construction gives $\text{tree}(G) \geq 1 + \lambda(G)$. When $g=3$, the conjectured right-hand side is

$$\lfloor 3/2 \rfloor - 1 + \lambda(G) = \lambda(G),$$

so the claim follows.

4.3. Girth at least four

Assume $g \geq 4$. The graph is triangle-free, so every neighborhood is an independent set. Consequently

$$\ell(u) = d(u) \text{ for every } u, \lambda(G) = \Delta(G).$$

Choose a vertex v of maximum degree and set

$$r = \lfloor g/2 \rfloor - 1.$$

By Lemma 1, there is a vertex at distance at least r from v . Taking the first r edges of a shortest path to that vertex gives a geodesic

$$v = x_0, x_1, \dots, x_r.$$

Consider

$$U = v \cup N(v) \cup x_2, \dots, x_r.$$

We claim that $G[U]$ is a tree. The path $x_0 x_1 \dots x_r$ is induced, because a chord would shorten the path from v . The set $N(v)$ is independent. It remains to rule out an edge from a different star leaf $y \in N(v) \setminus x_1$ to the extended branch. If yx_i were an edge for some $2 \leq i \leq r$, then

$$v, y, x_i, x_{i-1}, \dots, x_1, v$$

would be a cycle of length $i+2$. But

$$i+2 \leq r+2 = \lfloor g/2 \rfloor + 1 < g,$$

a contradiction. Thus $G[U]$ is precisely a star centered at v with the branch through x_1 extended to the path $v, x_1, \dots, x_r$. It is connected and acyclic, hence an induced tree.

The only overlap between $N(v)$ and the geodesic is x_1 , so

$$|U| = 1 + d(v) + (r-1) = \Delta(G) + r = \lambda(G) + \lfloor g/2 \rfloor - 1.$$

Therefore $\text{tree}(G) \geq |U|$, proving the theorem. $\square$

For $g=4$ or 5 , one has $r=1$; the construction then reduces to the induced star itself. The path extension becomes nontrivial from girth six onward.

5. Lean formalization

The accompanying Lean 4 development uses Lean and mathlib version 4.27.0, matching the toolchain pinned by Formal Conjectures when this repository was prepared. Its final declaration is:

```
theorem Graffiti141.conjecture141
  (G : SimpleGraph  $\alpha$ ) [DecidableRel G.Adj] (hG : G.Connected) :
  (G.girth / 2 :  $\mathbb{Z}$ ) - 1 +
  ((Finset.univ.sup (indepNeighborsCard G) :  $\mathbb{N}$ ) :  $\mathbb{Z}$ )  $\leq$ 
  (largestInducedTreeSize G :  $\mathbb{Z}$ )
```

The formal proof mirrors the written construction. It first builds a breadth-first spanning tree by choosing, for each non-root vertex, a neighbor one distance layer nearer the root. It proves this graph is a tree and that it preserves all distances from the root. A non-tree edge then supplies the short cycle used in Lemma 1. The geodesic support is an induced tree; the remaining neighbors of the root are added one at a time as leaves, with the girth argument proving that the root is each leaf's unique neighbor in the current tree. Finally, a one-vertex intersection bound yields the cardinality calculation.

The repository contains no `sorry`, `admit`, or user-declared axiom. Lean's reported foundational dependencies of the final theorem are the standard `propext`, `Classical.choice`, and `Quot.sound` used throughout mathlib.

6. Related formalization and provenance

An independent formal proof of Conjectures 141 and 143 by the GitHub contributor AlperTheKing was submitted earlier as google-deepmind/formal-conjectures pull request #4454. That development proves Conjecture 141 through a stronger bound obtained by growing a maximum induced tree containing a closed neighborhood. The present proof instead uses the explicit maximum-degree star plus geodesic extension described above. We cite that prior formalization and make no claim of priority for the first Lean proof.

Two generic Lean lemmas in this repository---attaching a uniquely adjacent leaf to an induced tree and recognizing an induced star---are adapted under Apache-2.0 from that pull request and are identified in NOTICE. The BFS construction, geodesic-extension argument, and final proof in this repository implement the route of the present note.

References

1. E. DeLaVina, [*Written on the Wall II: Conjectures of Graffiti.pc*](#), Conjecture 141.
2. Google DeepMind, [*Formal Conjectures: GraphConjecture141*](#).
3. AlperTheKing, [“Prove WOWII Graph Conjectures 141 and 143,”](#) Formal Conjectures pull request #4454, 2026.
4. The mathlib Community, [*mathlib: a mathematical library for Lean*](#).